Graph Convolutional Optimal Transport for Hyperspectral Image Spectral Clustering

Shujun Liu, *Graduate Student Member, IEEE*, and Huajun Wang

***Abstract*— Suffering from rich spectral and spatial information, the hyperspectral images (HSIs) that embed low-dimensional nonlinear manifolds lead to a challenging clustering task. In this article, we propose a promising clustering method for HSIs, termed as graph convolutional optimal transport (GCOT). For capturing the intrinsic geometric structure of data, we develop optimal transport (OT) on their graph embeddings. The OT problem learns discrete transport coupling to form a natural affinity matrix used for spectral clustering. Different from most other methods of building a graph with the Euclidean distance among neighborhoods, we leverage the OT probability to seek the edges of the graph that characterizes the local nonlinear structure of the original feature. For profound understanding of the proposed model, we prove the coupling holds sparsity and low-rank property that are beneficial to spectral clustering and provide its relationship with subspace clustering. To the best of our knowledge, this work could be one of the first spectral clustering methods using the OT theory. Several experiments on three widely used datasets with subjective and objective evaluation show that the proposed method dramatically outperforms the common existing methods and yields a comparable result with the state-of-the-art methods. Especially, it is significantly superior to the latter ones on computational efficiency.**

***Index Terms*— Graph convolution, hyperspectral image (HSI) clustering, optimal transport (OT), subspace clustering.**

I. Introduction

HYPERSPECTRAL images (HSIs) captured by remote sensing satellites and aircraft have rich spectral resolution and large spatial scale, bringing to broad applications in geophysics, military, environment, and medicine [1], [2]. Recently, the lack of labeled training samples makes unsupervised HSIs clustering, which aims to segment pixels into different object categories, attracting rising attention [3], [4], [5].

Among the unsupervised clustering methods, the spectral-clustering-based methods have achieved great success [6], [7], [8], [9], in which the critical step is to construct a discriminative affinity matrix that is then fed to a spectral clustering algorithm to obtain clustering results. The most common approaches to seeking the affinity matrix are the subspace clustering methods, which solve for the linear self-expression coefficient matrix of data with different norm constraints. Formally, they optimize the representation matrix to form the affinity matrix by minimizing

$$\min_{\boldsymbol{\pi}} \mathcal{R}(\boldsymbol{\pi}) + \mathcal{L}(\mathbf{X} - \boldsymbol{\pi}\mathbf{X}) \tag{1}$$

where $\mathbf{X} \in \mathbb{R}^{N\times D}$ denotes the data matrix with samples of N and dimension of D, $\boldsymbol{\pi} \in \mathbb{R}^{N\times N}$ denotes the linear representation of the data for solving, $\mathcal{L}(\cdot)$ denotes the reconstruction error function, and $\mathcal{R}(\cdot)$ determines that each point is only expressed by other points that lie in the same subspace with it. For example, sparse subspace clustering (SSC) [10], low-rank representations [11], [12] and low-rank subspace clustering (LRSC) [13] adopt the ℓ_1-norm and ℓ_*-norm (nuclear norm) as $\mathcal{R}(\cdot)$ to generate sparse and low-rank representations $\boldsymbol{\pi}$. To characterize the global structure information of data, Laplacian-regularized low-rank subspace clustering (LLRSC) [14] adds a Laplacian regularization into the LRR problem. By leveraging various aspects of prior knowledge of data, spectral-spatial sparse subspace clustering (S^4C) [15], robust manifold matrix factorization (RMMF) [3], and L2-SSC [16] make some changes to the form of (1). These objective functions are usually optimized by the alternating direction method of multipliers (ADMM) [17]. To reduce the tremendous computational complexity caused by large HSIs, sketched subspace clustering (sketched SC) [18] solves an asymmetric representation on a sketched dictionary instead of the self-expression. Similarly, SCC [19] optimizes the sparse coefficient on a structured dictionary, but generates clusters by minimum recovery residual rather than the spectral clustering. Recently, some HSI clustering methods with different mechanisms have also achieved appealing results [20], e.g., affinity propagation based on structural similarity and local outlier factor (CLAP) [21], and affinity matrix learning via nonnegative matrix factorization (NMFAML) [22].

However, suffering from the Hughes phenomenon [23], mixed pixels, imaging mechanisms, and atmospheric conditions during acquisition [1], the HSIs usually form nonlinear structure that no longer satisfies the linear condition required by subspace clustering. To alleviate this problem, one way is to map the original data into high-dimensional linear separable space by applying the kernel trick, which has been attempted by [24], [25], [26], [27]. However, these approaches based on the kernel Gram matrix require an elaborate selection of kernel functions and usually do not make geometric sense. Another workarounds are attributed to deep learning models, the basic ideas of which are similar to kernel mapping, transforming the data $\mathbf{X}$ into hidden features $\phi(\mathbf{X})$

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Shujun Liu is with the Key Laboratory of Earth Exploration and Information Techniques of Ministry of Education, College of Geophysics, Chengdu University of Technology, Chengdu 610059, China (e-mail: suldier@outlook.com).

Huajun Wang is with the College of Mathematics and Physics, Chengdu University of Technology, Chengdu 610059, China (e-mail: wanghuajun@cdut.edu.cn).

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to capture the inherent structure, where $\phi(\cdot)$ denotes some certain neural networks. To apply the deep model for the HSIs' subspace clustering, several literature [28], [29], [30], [31], [32] introduce the autoencoder architectures into the subspace clustering framework, resulting in more linear representation of data. Although deep-model-based methods produce more significant performance than the conventional methods, they still rely on linear multiplication forms in optimization objects, where the hidden features may not satisfy the closure of the linear subspace definition. Besides, they often lack a strong theoretical explanation for the intuition of network building.

Meanwhile, over the past decade, optimal transport (OT) has been used successfully in numerous clustering tasks [33], [34], [35], [36], owing to the rapid development of a complete theoretical foundation [37], [38], [39], [40] and efficient optimization algorithms [41], [42], [43]. Considering two piles of data with ground distance or cost, e.g., Euclidean distance, geodesic distance, and divergence, OT defines the least distance between two probability distributions that the data follow. For discrete OT problem, its solution, which is a optimal coupling matrix, indicates a correlation between samples (also referred to as soft alignment). Note that the coupling is also known as the discrete probability distribution, and this OT problem is also termed as the Kantorovich problem [44]. Based on the principle that the optimal coupling values are inversely proportional to the distances between their corresponding sample positions, the heterogeneous domain adaptation (HDA) tasks of Gromov-Wasserstein (GW) [45] and CO-optimal transport (COOT) [46] have shown that the optimal coupling learned from two kinds of data with aligned objects can form a diagonal block structure, where each block represents the correspondence between samples in a cluster. The minimum OT distance between two homologous data will be close to zero if the transport cost is a distance, which means the resulting coupling has converged to a sparse solution, but not to one that has the diagonal block structure. To fix this drawback, entropically regularized OT [43], [47] adds an entropy regularization to the basic OT, yielding more diffuse paths for optimal coupling.

Inspired by the previous analysis, we consider developing the affinity matrix of spectral clustering on OT coupling, which is obtained by minimizing the OT distance with entropic regularization between HSIs' data. Unfortunately, the ground distance of OT in the spatial domain of HSIs often remains a linear relationship. As a result, optimal coupling is flawed in capturing their intrinsic structure. To circumvent this limitation, graph convolutional networks (GCNs) [48] and their variants [49], [50], [51] developed on convolutional neural networks (CNNs) provide an effective scheme. By performing convolution on graphs, GCNs learn graph representations that endow the features in the Euclidean space with graph structure.

In this work, we propose a novel clustering method for HSIs' data, by introducing OT into a GCN, abbreviated as graph convolutional OT (GCOT). It measures the OT distance between the graph embeddings of HSIs' data that are endowed with uniform distribution and captures the intrinsic geometric structure of data. Significantly, unlike most graph convolutions [52], [53] that construct the adjacency matrix of the graph according to the nearest k neighbors, we achieve this by choosing the top k transport probabilities of OT coupling for each point. Hence, the new adjacency matrix that indicates the connectivity of data is equivalent to its optimal transformation paths. To the best of our knowledge, GCOT could be the first generalization of OT exploiting graph convolution and using it for HSIs' clustering based on spectral clustering. In summary, the main contributions of this article are highlighted as follows.

1) A promising OT variant using graph convolution, GCOT, is proposed for HSIs' clustering, where the new adjacency graph constructed by the optimal transformation path enforces the graph convolution to produce robustness graph embedding.
2) OT coupling is introduced to construct the Laplacian matrix for improving spectral clustering.
3) Theoretical proofs of sparsity and low-rank property of GCOT coupling and the relationship with subspace clustering are given for interpretability.
4) Several experiments are conducted on three HSI datasets, showing significant performance of the proposed method in terms of clustering effectiveness and efficiency.

The remainder of this article is organized as follows. A brief review of OT and convolutional networks (GCNs) is provided in Section II. In Section III, we present a detailed description of the proposed GCOT and its theoretical analysis. The experimental results and discussion are shown in Section IV, and the conclusion and future outlook are given in Section V.

II. Related Work

A. Notations

The probability simplex with N bins is written as $\Sigma_N = \{\mathbf{a} \in \mathbb{R}_+^N : \sum_{i=1}^N \mathbf{a}_i = 1\}$. For discrete distributions $\alpha = \sum_{i=1}^N \mathbf{a}_i \delta_{\mathbf{x}_i}$, $\beta = \sum_{i=1}^N \mathbf{b}_i \delta_{\mathbf{y}_i}$ supported on $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N \in \mathcal{X}$ and $\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_N \in \mathcal{Y}$, $\mathbf{a} \in \Sigma_{N_1}$, and $\mathbf{b} \in \Sigma_{N_2}$, the set of coupling measures between α and β reads $\mathcal{U}(\alpha, \beta) = \{\boldsymbol{\pi} \in \mathbb{R}_+^{N_1 \times N_2} : \boldsymbol{\pi} \mathbb{1}_{N_2} = \mathbf{a}, \boldsymbol{\pi}^\top \mathbb{1}_{N_1} = \mathbf{b}\}$. Note that $\mathbb{1}_N \in \mathbb{R}^N$ denotes an N-dimensional vector of ones, and $\mathbb{I}_N$ denotes an identity matrix with size of $N \times N$. The Frobenius dot product for two matrices is written as $\langle \cdot, \cdot \rangle$, i.e., $\langle \mathbf{X}, \mathbf{Y} \rangle = tr(\mathbf{X}^\top \mathbf{Y})$ where $tr(\cdot)$ stands for the trance of a matrix. We further define a graph as $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{A})$ that contains the notes of $|\mathcal{V}| = N$, the edge set $\mathcal{E}$, and their corresponding adjacency matrix $\mathbf{A} \in \mathbb{R}^{N \times N}$. Finally, we write scalars in lower case Greek letters or lower case letters (e.g., ϵ, k), vectors in bold lower-case letters (e.g., $\mathbf{x}$), and matrices in bold upper case letters (e.g., $\mathbf{X}$) or bold lower case Greek letters (e.g., $\boldsymbol{\pi}$) throughout the article.

B. Optimal Transport

OT distance provides a powerful geometric way to measure the distance between two probability measures or distributions. Given data samples $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N] \in \mathcal{X}$ ($\mathbb{R}^{N_1 \times D}$) and $\mathbf{Y} = [\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_N] \in \mathcal{Y}$ ($\mathbb{R}^{N_2 \times D}$) subjecting to probability distributions $\alpha = \sum_{i=1}^{N_1} \mathbf{a}_i \delta_{\mathbf{x}_i}$ and $\beta = \sum_{i=1}^{N_2} \mathbf{b}_i \delta_{\mathbf{y}_i}$, where the weights $\mathbf{a} \in \Sigma_{N_1}$ and $\mathbf{b} \in \Sigma_{N_2}$ are defined on probability simplexes Σ_{N_1} and Σ_{N_2}, and $\delta_{\mathbf{x}_i}$ indicates the Dirac at position $\mathbf{x}_i$, the entropically regularized OT problem [43], [47]

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with a certain transportation cost $\mathbf{C} = (c(\mathbf{x}_i, \mathbf{y}_j)_{i,j}) \in \mathbb{R}_+^{N_1 \times N_2}$ from α to β is written as

$$OT_\epsilon(\alpha, \beta) = \min_{\boldsymbol{\pi} \in \mathcal{U}(\alpha,\beta)} \langle \mathbf{C}, \boldsymbol{\pi} \rangle - \epsilon \mathbf{H}(\boldsymbol{\pi}) \tag{2}$$

$$\mathbf{H}(\boldsymbol{\pi}) = \sum_{i,j} -\boldsymbol{\pi}_{i,j}(\log(\boldsymbol{\pi}_{i,j}) - 1). \tag{3}$$

Here, $\boldsymbol{\pi}$ denotes the transport coupling matrix, $\boldsymbol{H}(\boldsymbol{\pi})$ denotes the entropy of $\boldsymbol{\pi}$, and ϵ denotes an adjustable threshold that affects the optimal coupling position in solution space and its sparsity degree [33]. It is well-known that $OT_\epsilon(\alpha, \beta)$ is equivalent to the 1-Wasserstein distance [54] when the sample sizes of $\mathbf{X}$ and $\mathbf{Y}$ are equal and $\mathbf{C}$ is a distance.

This OT problem can be efficiently optimized by the Sinkhorn algorithm [43]. Specifically, the transforming problem of (2) into a dual problem is solved by alternately updating

$$\mathbf{u}^{(t+1)} = \frac{\mathbf{a}}{\mathbf{K}\mathbf{v}^{(t)}}, \quad \mathbf{v}^{(t+1)} = \frac{\mathbf{b}}{\mathbf{K}^\top \mathbf{u}^{(t+1)}} \tag{4}$$

until convergence, where $\mathbf{K} = e^{-\mathbf{C}/\epsilon}$, $\mathbf{u}$, and $\mathbf{v}$ are temporary variables initialized with random positive vectors. As a result, optimal coupling is achieved by

$$\boldsymbol{\pi} = \mathbf{u}\mathbf{K}\mathbf{v}^\top. \tag{5}$$

Because the optimizer only involves multiplication and elementwise division operators, it can be optimized with GPU acceleration [55]. Once we obtain OT coupling, which indicates the transmission mode between samples, the clustering results are achieved by implementing the spectral clustering algorithm based on the coupling matrix.

C. Graph Convolutional Networks

Graph convolution makes the convolution operator on the graph domain available, derived from the Fourier analysis on the graph originally. The Fourier transform on the graph is equivalent to eigendecomposition of its Laplacian matrix [56]. That is, the Fourier bases correspond to the eigenvectors and the frequencies of the graph correspond to the eigenvalues. Thanks to the convolution theorem, the convolution operator acting between the function and the filter on the graph can be expressed as the inverse Fourier transform of multiplication between their Fourier transforms, so these kinds of graph convolutions are also called as spectral convolutions. Based on this route, GCNs approximate the spectral convolution by adopting the 1th order polynomials of the Laplacian matrix, changing its largest eigenvalue to 2, and adjusting the coefficients of the eigenvalues. For numerical stability of the graph convolution applied in deep networks, GCN normalizes graph by the renormalization trick [48], getting

$$\begin{cases} \mathbf{H}^k = \sigma\left(\overline{\mathbf{H}}^{(k)}\mathbf{W}^{(k)}\right) \\ \overline{\mathbf{H}}^{(k)} = \mathbf{S}\mathbf{H}^{(k-1)} \\ \mathbf{S} = \widetilde{\mathbf{D}}^{-1/2}\widetilde{\mathbf{A}}\widetilde{\mathbf{D}}^{-1/2} \end{cases} \tag{6}$$

where $\widetilde{\mathbf{A}} = \mathbb{I}_N + \mathbf{A}$ is an adjacency matrix having added self-loops, $\widetilde{\mathbf{D}} = \text{diag}(\widetilde{\mathbf{A}}\mathbb{1}_N)$ is the degree matrix corresponding to $\widetilde{\mathbf{A}}$, $\mathbf{W}$ is a learnable weight matrix of the network, and σ denotes an activation function. Generally speaking, each graph convolution layer of the stacked GCN in (6) consists of nonlinear activation (i.e., $\sigma(\cdot)$), linear transformation ($\overline{\mathbf{H}}^{(k)}\mathbf{W}^{(k)}$), and feature propagation (i.e., $\mathbf{S}\mathbf{H}^{(k-1)}$) [57], which is considered as a smooth operator on local graph along the edges and also as the graph gather layer in graph neural networks (GNNs) [58].

III. GCOT for Clustering

In this section, we first present the details of our proposed method, GCOT, and then present HSI clustering using GCOT. Finally, we provide proofs of the optimal coupling properties of GCOT and its relationship with other clustering methods.

A. GCOT

To capture the intrinsic nonlinear structure of data, we extend OT using graph convolution, in which the graph is constructed by optimal coupling. Specifically, we consider a data matrix $\mathbf{X} \in \mathbb{R}^{N \times D}$ with points of N and dimensions of D. Suppose the empirical distribution of $\mathbf{X}$ is α defined by weight $\mathbf{a} \in \Sigma_N$. Note that if the distribution of data is unknown, uniform distribution is often considered as an appropriate prior for it. We write the proposed GCOT problem as

$$\text{GCOT}_\epsilon(\alpha, \beta) = \min_{\boldsymbol{\pi} \in \mathcal{U}(\alpha,\beta)} \langle \mathbf{C}(\widetilde{\boldsymbol{\pi}}\mathbf{X}, \widetilde{\boldsymbol{\pi}}\mathbf{X}), \boldsymbol{\pi} \rangle - \epsilon \mathbf{H}(\boldsymbol{\pi}) \tag{7}$$

where $\mathbf{C}(\cdot,\cdot)$, $\mathbf{H}(\cdot)$, and $\boldsymbol{\pi}$ denote the cost matrix, entropic regularizer, and transport coupling matrix, respectively. Significantly, we represent the adjacent matrix of the graph of data as $\widetilde{\boldsymbol{\pi}}$, where $\widetilde{\boldsymbol{\pi}}_{i,j}$ is calculated by the transport probability from $\mathbf{x}_i$ to $\mathbf{x}_j$, different from most other methods based on the Euclidean distance between them. Formally, the adjacent matrix is defined as

$$\widetilde{\boldsymbol{\pi}} = \widetilde{\mathbf{D}}^{-1/2}\widehat{\boldsymbol{\pi}}\widetilde{\mathbf{D}}^{-1/2} \tag{8}$$

$$\widehat{\boldsymbol{\pi}}_{i,j} = (\mathbb{I}_\mathbb{N})_{i,j} + \chi_{\mathcal{A}_i^k}(\boldsymbol{\pi}'_{i,j})\chi_{\mathcal{A}_i^k}\left(\boldsymbol{\pi}'^\top_{i,j}\right) \tag{9}$$

$$\widetilde{\mathbf{D}} = \text{diag}(\widehat{\boldsymbol{\pi}}\,\mathbb{1}_N) \tag{10}$$

where $\widetilde{\mathbf{D}}$ denotes the normalization term, $\mathcal{A}_i^k$ denotes a set that consists of the top k values in the kth row of the OT coupling matrix $\boldsymbol{\pi}'$ on $\mathbf{X}$, and $\chi_\mathcal{A}(x)$ denotes an indicator function formulated as

$$\chi_\mathcal{A}(x) = \begin{cases} 1, & x \in \mathcal{A} \\ 0, & \text{otherwise.} \end{cases} \tag{11}$$

Obviously, $\widetilde{\boldsymbol{\pi}}$ is symmetric and contains a graph structure with self-connection. As a result, the goal of problem (7) is to seek the OT coupling matrix $\boldsymbol{\pi}$ in the set of coupling measures $\mathcal{U}(\alpha, \beta)$.

Before implementing HSI clustering via GCOT, we would like to give an insight into the cost matrix $\mathbf{C}(\widetilde{\boldsymbol{\pi}}\mathbf{X}, \widetilde{\boldsymbol{\pi}}\mathbf{X})$ from manifold viewpoint. Suppose data points $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N \in \mathbb{R}^D$ lie on a d-dimensional manifold $\mathcal{M}$, where $d < D$. It is well-known that $\mathcal{M}$ is locally homeomorphic with a low-dimensional Euclidean space $\mathbb{R}^d$. That is, every $\mathbf{y} \in \mathcal{M}$ has an open neighborhood $\mathcal{N}_\mathbf{y}$ and there exists a homeomorphism $f : \mathcal{N}_x \to \mathbb{R}^d$, which is a continuous function whose inverse function is also continuous [59]. As mentioned in [27], the graph convolution $\widetilde{\boldsymbol{\pi}}\mathbf{X}$ is a special case of

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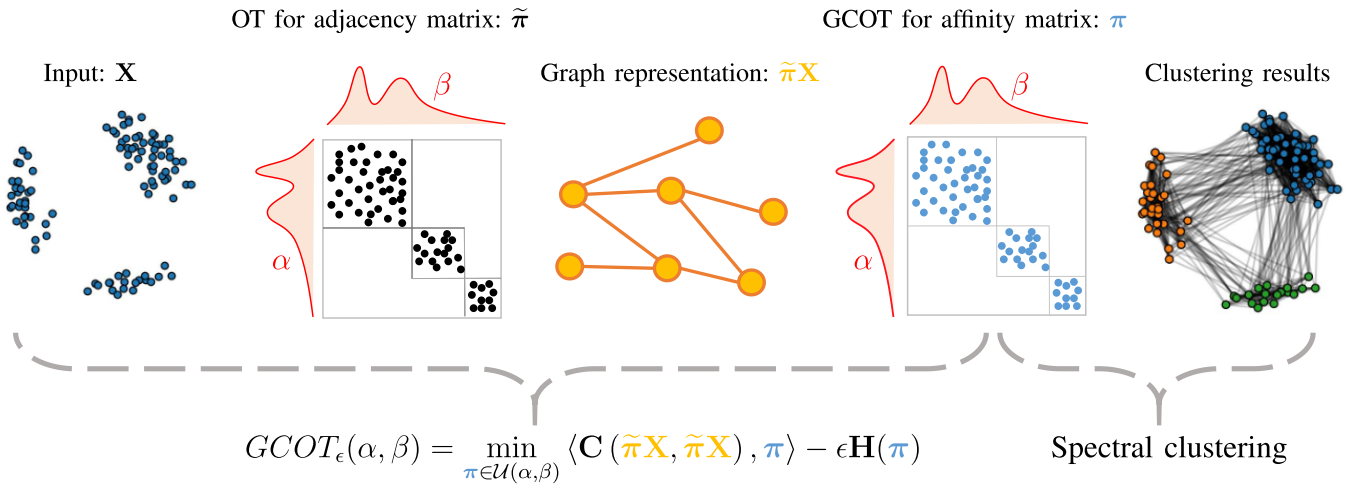


Fig. 1. Overview of the proposed GCOT for clustering. For a given data $\mathbf{X}$ assigned with distributions α and β, we first construct the adjacency matrix $\widetilde{\pi}$ on the optimal coupling of OT from $\mathbf{X}$ to itself. To characterize the intrinsic geometry of data, we then map $\mathbf{X}$ to graph representation, $\widetilde{\pi}\mathbf{X}$, by implementing graph convolution with graph $\widetilde{\pi}$. Finally, we optimize the OT problem on the graph embedding $\widetilde{\pi}\mathbf{X}$, leading to optimal coupling π whose sparsity and low-rank property are controlled by ϵ. The result clusters are obtained by applying spectral clustering to the coupling matrix.

Laplacian smoothing [60]. Each point $\mathbf{x} \in \widetilde{\pi}\mathbf{X}$ is linearly represented by its adjacent nodes on the graph with OT distance. It actually treats a local manifold embedded in the space of points $\mathbf{X}$ as linear, leading to more discriminative feature distribution in the graph domain. Therefore, $\widetilde{\pi}\mathbf{X}$ contains the local geometric information of $\mathbf{X}$. Because $\mathbf{C}(\widetilde{\pi}X, \widetilde{\pi}X)$ is calculated by the distances or divergences between all the points of $\widetilde{\pi}\mathbf{X}$, the cost matrix $\mathbf{C}$ contains both the local and global geometric structures of $\mathbf{X}$. Such cost matrix that has characterized the intrinsic structure of data promotes the nonlinear representation ability of the transport coupling in the original OT problem.

The general clustering scheme using GCOT for HSIs is illustrated in Fig. 1, where the clustering program can be divided into two steps, forming an affinity matrix by GCOT and applying spectral clustering on the affinity matrix. Specifically, we reshape 3-D HSIs' data as a matrix $\mathbf{X} \in \mathbb{R}^{N\times D}$ with samples' number of N and dimension of D and endow it with a uniform distribution. For building the graph on $\mathbf{X}$, we minimize the OT problem from $\mathbf{X}$ to itself to obtain an optimal coupling matrix, on which we then construct an adjacent matrix $\widetilde{\pi}$ representing a graph via (8)–(11). By applying the graph convolution operator, we subsequently get graph embedding of data, $\widetilde{\pi}\mathbf{X}$. Based on the graph embedding, we further optimize problem (7) to produce the OT coupling, π. As a result, a symmetric affinity matrix is formed by $\mathbf{A} = \pi + \pi^{\top}$, which is supplied to the spectral clustering algorithm to produce clustering results. To sum up, the complete process using the proposed method for HSIs' clustering is highlighted in Algorithm 1. The algorithm of GCOT for clustering consists of four steps: 1) calculating the distance matrix $\mathbf{C}'$ of the original features $\mathbf{X}$ and getting the optimal coupling π' by performing the Sinkhorn algorithm on $\mathbf{C}'$; 2) constructing adjacent matrix $\widetilde{\pi}$ and computing the graph embedding $\mathbf{H}$; 3) obtaining the optimal coupling π for the graph embedding similar to (1); and 4) forming the affinity matrix and applying the spectral clustering on it to produce clusters.

Algorithm 1 GCOT for Clustering

Input: Data $\mathbf{X}$, thresholds for OT ϵ_1 and ϵ_2, top k values of each raw and column of transport coupling, c clusters.
Initialize: $\pi_1 = \mathbb{1}_N$, $\mathbf{a} = [1/N, \ldots, 1/N]^{\top}{}_N$.
//Calculate distance matrix.
$\mathbf{C}' = \mathrm{diag}\left(\mathbf{X}\mathbf{X}^{\top}\right)\mathbb{1}_N^{\top} + \mathbb{1}_N\,\mathrm{diag}\left(\mathbf{X}\mathbf{X}^{\top}\right) - 2\mathbf{X}\mathbf{X}^{\top}$
//Sinkhorn algorithm.
$\mathbf{v}' = \mathbb{1}_N$, $\mathbf{K}' = e^{-\mathbf{C}'/\epsilon_1}$
while (not converge)
  $\mathbf{u}' = \mathbf{a}'/\mathbf{K}'\mathbf{v}'$, $\mathbf{v}' = \mathbf{a}'/\mathbf{K}'^{\top}\mathbf{u}'$
end while
$\pi' = \mathrm{diag}(\mathbf{u}')\mathbf{K}'\,\mathrm{diag}(\mathbf{v}')$
Build adjacency matrix $\widetilde{\pi}$ from π' by Eq. (8) - (11) with k.
Perform graph convolution, $\mathbf{H} = \widetilde{\pi}\mathbf{X}$.
//Calculate distance matrix.
$\mathbf{C}'' = \mathrm{diag}\left(\mathbf{H}\mathbf{H}^{\top}\right)\mathbb{1}_N^{\top} + \mathbb{1}_N\,\mathrm{diag}\left(\mathbf{H}\mathbf{H}^{\top}\right) - 2\mathbf{H}\mathbf{H}^{\top}$
//Sinkhorn algorithm.
$\mathbf{v} = \mathbb{1}_N$, $\mathbf{K} = e^{-\mathbf{C}''/\epsilon_2}$
while (not converge)
  $\mathbf{u} = \mathbf{a}/\mathbf{K}\mathbf{v}$, $\mathbf{v} = \mathbf{a}/\mathbf{K}^{\top}\mathbf{u}$
end while
$\pi = \mathrm{diag}(\mathbf{u})\mathbf{K}\,\mathrm{diag}(\mathbf{v})$
end while
Perform spectral clustering on $\mathbf{A} = \pi + \pi^{\top}$ with c.
Output: Clustering results.

Complexity: Owing to the Sinkhorn algorithm, the optimizer of the OT problem converges quickly because of involving only alternate multiplication and elementwise division operators with a computational consumption of $O(N^2)$ [61]. Besides, as shown in Algorithm 1, the additional calculation consumption prominently derives from building two distance matrices and an adjacent matrix. To efficiently solve them requires $O(2N^3 + 2N\log(N))$ time complexity, i.e., matrix multiplication of N^3 and quick sort of $N\log(N)$ for constructing the graph.

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B. Optimal Coupling Properties of GCOT

For giving theoretical guarantee of clustering using GCOT, we would like to discuss two important properties of optimal coupling, i.e., $\boldsymbol{\pi}$ in problem (7). Most spectral-clustering-based methods cannot prevent affinity matrices with all-zero rows or columns so that the lost connectivity would cause unreasonable clustering results [62]. In contrast, the optimal transport coupling is a natural well-normalized affinity matrix for spectral clustering, because the solution of the GCOT problem belongs to the set of coupling measures $\mathcal{U}(\alpha, \beta) = \{\boldsymbol{\pi} \in \mathbb{R}_+^{N_1 \times N_2} : \boldsymbol{\pi} \mathbb{1}_{N_2} = \mathbf{a}, \boldsymbol{\pi}^\top \mathbb{1}_{N_1} = \mathbf{b}\}$, which constrains all rows' sum and columns' sum of each coupling to be distribution weights. In addition, when adopting uniform distribution, all rows and columns of the coupling matrix sums satisfy greater than zero, consequently without requiring the operation of absolute value to form affinity matrix.

As noted in the introduction, the affinity matrix with sparsity or low-rank property has been verified effective for spectral clustering, achieved by a considerable amount of work exploiting matrix ℓ_1- and ℓ_*-norms as constraints. Fortunately, the OT coupling matrix has both sparsity and low-rank property, thanks to the entropic regularization of OT, which is proven in the following, where, for convenience, we abbreviate the graph embedding $\widetilde{\boldsymbol{\pi}}\mathbf{X}$ of problem (7) to $\mathbf{X}$.

1) Sparsity: The threshold ϵ of the entropic regularization in (7) is to trade off between the bas OT distance and the divergence degree of transport paths. Based on that, here, we prove the OT coupling matrix is sparse if there are certain distances between different data clusters, as seen in Theorem 1.

Theorem 1: Considering a collection of data points $\mathbf{X} \in \mathbb{R}^{N \times D}$ sampled from n clusters, $\mathbf{X}_{+i} \in \mathbf{X}$ belongs to the ith cluster with N_i data samples, and $\mathbf{X}_{-i}$ comes from the remaining $n-1$ clusters. For a sample $\mathbf{x} \in \mathbf{X}_{+i}$, suppose $\mathbf{c}$ denotes the distance vector between $\mathbf{x}$ and $\mathbf{X}$, and $\mathbf{t}$ denotes the corresponding coupling vector. There, $\exists \epsilon \in (0, \infty)$, and OT optimization

$$\begin{bmatrix} \mathbf{t}_+^* \\ \mathbf{t}_-^* \end{bmatrix} = \min_{\mathbf{t} \in \mathcal{U}(\alpha,\beta)} \left\langle \begin{bmatrix} \mathbf{c}_+ \\ \mathbf{c}_- \end{bmatrix}, \begin{bmatrix} \mathbf{t}_+ \\ \mathbf{t}_- \end{bmatrix} \right\rangle - \epsilon \mathbf{H}(\mathbf{t}) \tag{12}$$

recovers a sparse transportation for every $\mathbf{x} \in \mathbf{X}_{+i}$, i.e., $\mathbf{t}_+{}^* \neq 0$ and $\mathbf{t}_-^* = 0$, if and only if $\min(\mathbf{c}_+) > \max(\mathbf{c}_-)$ holds.

Proof: (Sufficiency) Without loss of generality, we assume that the elements of $\mathbf{c} = [\mathbf{c}_+^\top, \mathbf{c}_-^\top]^\top$ are arranged in ascending order of size. As shown in Proposition 4.1 in [33], when ϵ goes to 0, the OT distance of (12) converges to its nonregularized version, i.e., $\epsilon = 0$. Hence, the optimal solution of (12) with $\epsilon = 0$ is $[1, 0, \ldots, 0]$ with $N-1$ zero terms. On the other hand, according to the convex optimization theory, we know that the minimum of $-\mathbf{H}(\mathbf{t})$ is $[1/N, 1/N, \ldots, 1/N]$ without any nonzero terms. When ϵ goes to $+\infty$, the optimal solution of (12) converges a nonzero vector. Furthermore, since the $-\mathbf{H}(\mathbf{t})$ is continuous, there exists $\epsilon \in (0, \infty)$, such that the optimal solution of $-\mathbf{H}(\mathbf{t})$ is a N_i nonzero term transportation $\mathbf{t}^* = [\mathbf{t}_+^{*\top}, \mathbf{t}_-^{*\top}]$ that we arrange in descending order. As a result, $\langle [\mathbf{c}_+^\top, \mathbf{c}_-^\top], \mathbf{t}^* \rangle$ equals 0, and $\mathbf{t}^*$ is the sparse transportation from $\mathbf{x} \in \mathbf{X}_{+i}$ to $\mathbf{X}$.

(Necessity) Exploiting contradiction, we assume there is a $\mathbf{c}_{-j} \in \mathbf{c}_-$, $\mathbf{c}_{-j} > \min(\mathbf{c}_+)$, which indicates that the OT distance corresponding to $\mathbf{c}$ is smaller than one that $\mathbf{t}^*$ leads to. Hence it contradicts the sparse transportation assumption. ■

2) Low-Rank Property: Similar to the sparsity of the OT coupling matrix, we prove that the solution of the entropically regularized OT is low-rank, as seen in Theorem 2.

Theorem 2: Suppose $\boldsymbol{\pi}_\epsilon$ is an optimal solution of the OT problem (7) with maximal entropic regularization, where the distributions α with weight $\mathbf{a} \in \mathbb{R}_+^N$ and β with weight $\mathbf{b} \in \mathbb{R}_+^N$ are uniform and i.i.d.. Then $\exists \epsilon \in (0, \infty)$, $\boldsymbol{\pi}_\epsilon$ satisfies low-rank if $\mathbf{a}_i \neq 1$.

Proof: The convergence with ϵ of the optimal solution of (7) shows that [33]

$$\lim_{\epsilon \to 0} \boldsymbol{\pi}_\epsilon = \boldsymbol{\pi}_0, \quad \lim_{\epsilon \to +\infty} \boldsymbol{\pi}_\epsilon = \mathbf{a}\mathbf{b}^\top. \tag{13}$$

Since α and β are i.i.d., the left side of (13) implies $\boldsymbol{\pi}_0 = \operatorname{diag}(\mathbf{a})$ and the right side of (13) implies $\boldsymbol{\pi}_\infty = \mathbf{a}\mathbf{a}^\top$. Apparently

$$\operatorname{rank}(\boldsymbol{\pi}_0) = \|\mathbf{a}\|_0 = N, \quad \operatorname{rank}(\boldsymbol{\pi}_\infty) = 1 \tag{14}$$

where the rank function is discontinuous and nonconvex. Instead, we consider its tightest convex relaxation [63], i.e., the nuclear norm. The singular values of the diagonal matrix $\boldsymbol{\pi}_0$ together with the normalization of probability distribution imply that the left side of (14) can be converted into

$$\|\boldsymbol{\pi}_0\|_* = \|\mathbf{a}\|_1 = 1. \tag{15}$$

It is well-known that in singular value decomposition (SVD), the left nonzero singular values of a matrix are equal to its right nonzero singular values. As for the right side of (13), we can only consider the SVD of $\mathbf{a}^\top \mathbf{a}$

$$\mathbf{a}^\top \mathbf{a} = \mathbf{U}\Sigma\mathbf{V}^\top \tag{16}$$

where $\mathbf{U}$ and $\mathbf{V}$ are the real orthogonal matrices, and Σ is singular value diagonal matrix of rank of 1. Apparently, $\mathbf{a}^\top \mathbf{a}$ is a feasible singular value of Σ. Hence, we have the SVD of $\mathbf{a}^\top \mathbf{a}\mathbf{a}^\top \mathbf{a}$

$$(\mathbf{a}^\top \mathbf{a})^\top \mathbf{a}^\top \mathbf{a} = \mathbf{V}\Sigma\mathbf{U}^\top \mathbf{U}\Sigma\mathbf{V}^\top = \mathbf{V}\Sigma^2\mathbf{V}^\top \tag{17}$$

which implies Σ is the nonzero singular value of $\boldsymbol{\pi}_\infty$. Let $\sigma_1, \sigma_2, \ldots, \sigma_N$ be the singular value of $\mathbf{a}\mathbf{a}^\top$. Then, we have

$$\|\boldsymbol{\pi}_\infty\|_* = \sum_{i=1}^N \sigma_i = \mathbf{a}^\top \mathbf{a} + 0 + \ldots + 0 = \sum_{i=1}^N \mathbf{a}_i^2 \leq 1 \tag{18}$$

where $\|\boldsymbol{\pi}_\infty\|_* = 1$ is achieved if and only if $\mathbf{a}_i \neq 1$ and $\mathbf{a}_{j \neq i} = 0$. As a result, combining (14) and (18), we have

$$\|\boldsymbol{\pi}_0\|_* \geq \|\boldsymbol{\pi}_\infty\|_*. \tag{19}$$

Since $\|\boldsymbol{\pi}_\epsilon\|_*$ is continuous, (19) shows that

$$\exists \epsilon \in (0, +\infty), \|\boldsymbol{\pi}_\infty\|_* \leq \|\boldsymbol{\pi}_\epsilon\|_* \leq \|\boldsymbol{\pi}_0\|_* \tag{20}$$

i.e., $\boldsymbol{\pi}_\epsilon$ is low-rank if $\mathbf{a}_i \neq 1$. ■

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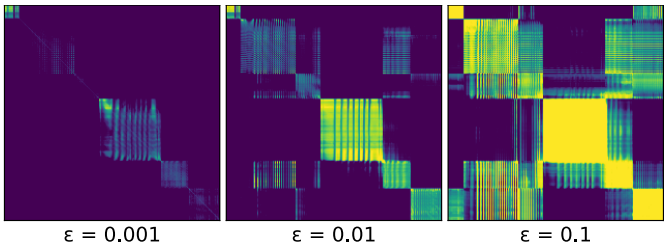


Fig. 2. Effects of ϵ on the OT coupling for the Salinas data set that contains six clusters. When we gradually increase ϵ, the coupling matrix gradients from sparse to divergent.

Note that in our proposal (7), through multiplication with the adjacency matrix, the graph convolution gathers the original feature notes into graph embedding prepared for downstream OT tasks. Without loss of generality, this concise graph map can be easily extended into various deep model versions with the learnable weights of multilayer neural networks, and the properties of OT coupling on the deep graph embedding hold as well as the former. In addition, to illustrate the effects of ϵ on the OT matrix $\boldsymbol{\pi}$ of the entropically regularized OT problem, we optimize it on the Salinas data, with different parameters $\epsilon = 0.001, 0.01, 0.1$ as can be seen in Fig. 2. From left to right in the figure, the coupling matrix gradually becomes divergent as ϵ increases. Based on this observation, our goal is find a distinct diagonal block matrix with an appropriate ϵ for clustering. In contrast, extremely sparse or divergent coupling will lead to poor clustering accuracy due to intractable segmentation on its affinity matrix.

C. Relationship With Subspace Clustering

As mentioned in the introduction, the subspace clustering problems consist of a regularization item of representation and a reconstruction error item formally. We generalize them with the OT problems together into a unified form, which reads

$$\min_{\boldsymbol{\pi}} \mathcal{R}(\mathbf{C} \odot \boldsymbol{\pi}) + \epsilon \mathcal{L}(\boldsymbol{\pi}, \mathbf{X}) \tag{21}$$

where $\mathbf{C}$ is a weighting matrix, $\mathcal{R}(\cdot)$ denotes a certain norm of the weighted coefficient $\mathbf{C} \odot \boldsymbol{\pi}$, and $\mathcal{L}(\cdot, \cdot)$ denotes a function with respect to $\boldsymbol{\pi}$ and $\mathbf{X}$. Consequently, when setting $\mathbf{C} = \mathbb{1}$, $\mathcal{R} = \ell_1$, and $\mathcal{L} = \mathcal{F} = \|\mathbf{X} - \boldsymbol{\pi}\mathbf{X}\|_F^2$, we have the SSC problem

$$\min_{\boldsymbol{\pi}} \|\boldsymbol{\pi}\|_1 + \epsilon \|\mathbf{X} - \boldsymbol{\pi}\mathbf{X}\|_F^2 \tag{22}$$

and also have the LRR and LRSC problems if $\mathcal{R} = \ell_*$

$$\min_{\boldsymbol{\pi}} \|\boldsymbol{\pi}\|_* + \epsilon \|\mathbf{X} - \boldsymbol{\pi}\mathbf{X}\|_F^2. \tag{23}$$

In addition to these subspace clustering problems, we briefly write the OT problem (7)

$$\min_{\boldsymbol{\pi}} \langle \boldsymbol{\pi}, \mathbf{X} \rangle + \epsilon \mathbf{H}(\boldsymbol{\pi}) + |\boldsymbol{\pi}\, \mathbb{1}_N - \mathbf{a}|_\infty + \left|\boldsymbol{\pi}^\top \mathbb{1}_N - \mathbf{a}\right|_\infty \tag{24}$$

as

$$\min_{\boldsymbol{\pi}} \langle \mathbf{C}, \boldsymbol{\pi} \rangle + \epsilon \mathcal{H}(\boldsymbol{\pi}, \mathbf{X}). \tag{25}$$

Note that because of $\boldsymbol{\pi} \in \mathcal{U}(\alpha, \beta)$, we get $\boldsymbol{\pi} \geq 0$, and (25) is equivalent to

$$\min_{\boldsymbol{\pi}} \|\mathbf{C} \odot \boldsymbol{\pi}\|_1 + \epsilon \mathcal{H}(\boldsymbol{\pi}, \mathbf{X}). \tag{26}$$

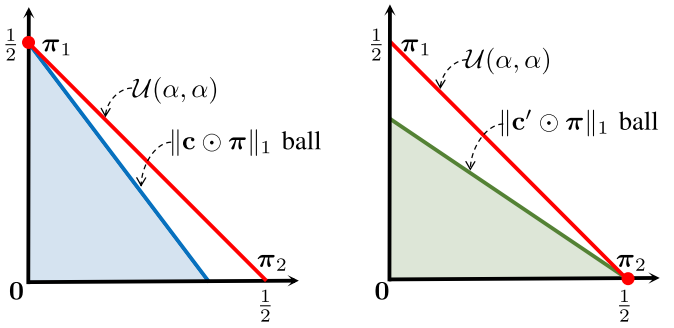


Fig. 3. OT problem is equivalent to a weighted ℓ_1-norm form with probability measure constraints $\mathcal{U}(\alpha, \alpha)$, where the weights $\mathbf{c}$ considered as the costs of OT affect the optimal solution.

So far, on one hand, the OT distance has another expression in the weighted ℓ_1-norm, $\|\mathbf{C} \odot \boldsymbol{\pi}\|_1$, as shown in (26). On the other hand, using the weighted ℓ_1 regularization in the compressed sensing (CS) task, Candès *et al.* [64] have achieved faster optimization and more effective sparse solution for signal recovery than the original CS program. Inspired by this, weighted sparse subspace clustering (WSSC) [65] introduces a weighting matrix calculated by the Euclidean distances between samples to the ℓ_1 regularization of the SSC framework. Weighting and kernel sparse subspace clustering (WKSSC) [66] to capture the local and global structures of data manifolds constructs a new weighting matrix using the similarities between tangent spaces in addition to the Euclidean distances on data manifolds. They lead to impressive clustering results and strong geometric interpretation, defined as the flowing unified form

$$\min_{\boldsymbol{\pi}} \|\mathbf{C} \odot \boldsymbol{\pi}\|_1 + \epsilon \mathcal{F}(\boldsymbol{\pi}, \mathbf{X}) \tag{27}$$

where $\mathcal{F}(\boldsymbol{\pi}, \mathbf{X})$ is the reconstruction error written as in (22). As a result, we observe that the OT problem (26) and WSSC problem (27) contain the same item, $\|\mathbf{C} \odot \boldsymbol{\pi}\|_1$, which indicates that the cost matrix of OT is equivalent to the weighting matrix of WSSC.

Hence, there is a new explanation of the effects of the cost matrix for the optimal transport coupling in OT problems from the weighted ℓ_1-norm aspect. Without loss of generality, here we consider a simplified OT example with 2 samples following a uniform distribution $\alpha = \sum_{i=1}^{2} \mathbf{a}_i \delta_{\mathbf{x}_i}$. Suppose $\boldsymbol{\pi}_*$ is a row of OT coupling $\boldsymbol{\pi}_{2\times 2} \in \mathcal{U}(\alpha, \alpha)$. Obviously, it satisfies $\sum_{i=1}^{2} \boldsymbol{\pi}_{*i} = 1/2$. Further assume $\boldsymbol{\pi}_1 = [1/2, 0]$ is the optimal solution of $\boldsymbol{\pi}_*$. As illustrated in Fig. 3, the blue ball denotes the weighted ℓ_1-norm $\|\mathbf{c} \odot \boldsymbol{\pi}_*\|_1$, and the red line denotes the solution space. With expanding the ball gradually from origin $\mathbf{0}$, it will first arrive at the point $\boldsymbol{\pi}_1$ on the red line, as shown in the left of the figure. However, when expecting that the solution of $\|\mathbf{c} \odot \boldsymbol{\pi}_*\|_1$ fast touches the point $\boldsymbol{\pi}_2 = [0, 1/2]$ than $\boldsymbol{\pi}_1$, the operation we should do is scaling up c_1 or scaling down c_2 in $\mathbf{c} = [c_1, c_2]$ and keeping another weight fixed, resulting in $\mathbf{c}'$ and the reweighted green ball $\|\mathbf{c}' \odot \boldsymbol{\pi}_*\|_1$ as drawn in the right of the figure. Significantly, while the cost matrix $\mathbf{C}$ of OT is seen as the original weights referring to the left case in the figure, the cost $\mathbf{C}'$ of GCOT is considered as its reweighted version referring to the right

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case. Theoretically, because the GCOT forces similar samples closer together than the original OT, its corresponding cost matrix tends to stretch the weighted ℓ_1 ball of GCOT into an appropriate shape conducive to subsequent clustering actually.

As well-known [10], when data $\mathbf{X} \in \mathbb{R}^{N\times D}$ lie in an affine subspace, an additional linear equality constraint should be added to the optimization objectives of the original SSC (21), rewritten as

$$\min_{\hat{\boldsymbol{\pi}}} \|\hat{\boldsymbol{\pi}}\|_1 + \epsilon\|\mathbf{X} - \hat{\boldsymbol{\pi}}\mathbf{X}\|_F^2 \quad \text{s.t.} \quad \hat{\boldsymbol{\pi}}\mathbb{1}_N = \mathbb{1}_N. \tag{28}$$

Let α and β be uniform distributions with weights $\mathbf{a}$ and $\mathbf{b}$. In the OT problem, since $\boldsymbol{\pi} \in \mathcal{U}(\alpha, \beta)$, we have a quality relationship between transport coupling $\boldsymbol{\pi}$ of OT and linear self-representation $\hat{\boldsymbol{\pi}}$ of SSC

$$\boldsymbol{\pi}\mathbb{1}_N = \mathbf{a} = 1/N\mathbb{1}_N = 1/N\hat{\boldsymbol{\pi}}\mathbb{1}_N. \tag{29}$$

IV. Experiments

A. Experimental Setup

1) Datasets and Preprocessing: To verify the effectiveness of the proposed method and facilitate comparison with other methods, we conduct comparative experiments on three widely used real scene HSI datasets, including Indian Pines, Salinas, and Pavia University [15], [30]. The Indian Pines dataset was collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor in Northwestern Indiana, USA. It is composed of 145 × 145 pixels and 220 spectral reflectance bands with a wavelength range of 0.4–2.5 μm and spatial resolution of about 20 m. After removing water absorption and noisy bands of 104–108, 150–163, and 220, the remaining 200 spectral bands are adopted in the experiments. The Salinas dataset was also acquired by the AVIRIS sensor over Salinas Valley in California, USA, with a spatial resolution of 3.7 m, a size of 512 × 217 pixels, and 224 bands. The water absorption bands of 108–112, 154–167, and 224 are neglected so that a total of 204 spectral bands are reserved for the experiments. The Pavia University dataset was captured by the Reflective Optics System Imaging Spectrometer (ROSIS-03) sensor at Pavia University, Italy, with a wavelength range of 0.43–0.86 μm and geometric resolution of 1.3 m. It is at a size of 610 × 340 pixels and with a final total number of 103 bands by discarding noisy bands of 12 from the original image.

For releasing the heavy computational burden, the typical subscenes are split from three original datasets within which the selected position are [30–115, 24–94], [591–676, 158–240], and [150–350, 100–200], respectively, as used in most work [4], [30], [53]. Furthermore, we transform the spectral band features into 4-D principal component features by applying principal component analysis (PCA), which preserves at least 95% of data information. It results in data shapes of 85 × 70 × 4, 83 × 86 × 4, and 140 × 150 × 4, and cluster numbers of 4, 6, and 8 for the datasets. Moreover, to capture both the spectral and spatial information [67], we shape the cube of neighboring pixels with a square form into a vector to represent its center point. Following in [27], we set the neighborhood size to be 9 for all the datasets, and scale all the feature values into [0, 1].

TABLE I
Parameter Settings of OT, GCOT-E, and GCOT on Three Datasets

Data sets	OT	GCOT-E		GCOT	
	ϵ	ϵ	k	ϵ	k
Indian Pines	0.3	0.25	20	0.32	20
Salinas	0.073	0.07	2	0.068	10
Pavia University	1.9	0.7	27	0.9	23

2) Evaluation Criteria: Three objective metrics that contain overall accuracy (OA), Kappa coefficient (Kappa), and normalized mutual information (NMI) [3], [30] are adopted for evaluating the clustering performance of all the comparison methods from different aspects. The OA metric directly calculates the accuracy according to the best alignment of the predicted clusters and ground-truth labels, while based on the correspondence the Kappa and NMI metrics provide assessments in the light of consistency check and mutual information. They all are normalized into [0, 1], where the clustering accuracy is considered to be monotonically increasing with the scaled score.

3) Comparison Methods and Parameter Settings: To evaluate the performance of our proposed method in terms of effectiveness and efficiency, we consider selecting several clustering methods in unsupervised manners, including LRR [12], LRSC [13], SSC [10], WSSC [65], S^4C [15], unsupervised broad learning (UBL) [4], RMMF [3], NMFAML [22], and essential graph convolutional subspace clustering (EGCSC) [27]. Among them, LRR, LRSC, and SSC belong to the conventional methods. Compared with our method, WSSC and EGCSC have similar methodologies. S^4C, UBL, RMMF, NMFAML, and EGCSC are the state-of-the-art methods in recent years.

For fair comparison, we stipulate all the methods use the same preprocessing scheme for all the datasets, described in detail above. For UBL, RMMF, and NMFAML, we show their results reported in the recent relevant literature [27], [53]. The hyperparameters of the remaining methods are found by implementing grid search to reach the best OA performance. Note that in the literature of S^4C, the untagged data of datasets are reserved but removed in our experiments, so we abandon its spatial constraint term from the optimization objective. We list the parameters of the proposed methods in Table I, where GCOT-E adopts the Euclidean distance instead of the OT probability in GCOT to build graph, and all the methods are used to ablation analysis. All the experiments are implemented on Intel Xeon Gold 5218 2.3-GHz CPU with 128-GB RAM. The source implementation of the proposed GCOT is released on the website.[1]

B. Evaluation Results

1) Objective Evaluation: The performance of all the methods on the Indian Pines, Salinas, and Pavia University datasets

[1] https://github.com/suldier/GCOT

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TABLE II
OBJECTIVE EVALUATION OF THE DIFFERENT CLUSTERING METHODS ON THE INDIAN PINES, SALINAS, AND PAVIA UNIVERSITY DATASETS WITH OA, KAPPA, AND NMI METRICS

Data sets	Metrics	Methods									
		LRR [12]	LRSC [13]	SSC [10]	WSSC [65]	S^4C [15]	UBL [4]	RMMF [3]	NMFAML [22]	EGCSC [27]	GCOT
Indian Pines	OA	0.6875	0.7611	0.6800	0.6914	0.5960	0.6258	0.7121	**0.8508**	0.8483	0.8404
	Kappa	0.5627	0.6534	0.5427	0.5750	0.4559	0.4690	0.5609	-	0.6422	**0.7654**
	NMI	0.4368	0.5007	0.5424	0.3682	0.3999	0.6680	0.4985	**0.7264**	0.6422	0.6258
Salinas	OA	0.8536	0.8485	0.7528	0.6421	0.8457	0.9142	0.9820	0.9956	**0.9985**	**0.9985**
	Kappa	0.8191	0.8135	0.6990	0.5625	0.8114	0.8943	0.9775	-	**0.9981**	**0.9981**
	NMI	0.7872	0.8029	0.7363	0.5881	0.7839	0.8692	0.9483	0.9848	**0.9949**	**0.9949**
Pavia University	OA	0.6932	0.7057	0.6458	0.5153	0.5564	0.7083	0.7704	0.8967	0.8442	**0.9102**
	Kappa	0.6128	0.6286	0.5520	0.4221	0.4777	0.6533	0.6804	-	0.7968	**0.8804**
	NMI	0.7016	0.7216	0.6680	0.3917	0.6885	0.6874	0.7388	0.9216	0.8401	**0.9233**

is shown in Table II, where "-" indicates no corresponding indicators reported in the related literature. On the Salinas and Pavia University datasets, the proposed GCOT obtains the highest scores of OA, Kappa, and NMI against the other HSI clustering methods. Especially, GCOT is superior to EGCSC that achieves best scores among all the subspace-clustering-based methods, with 8%, 10%, and 10% improvement in OA, Kappa, and NMI, respectively, and also better than the latest method of NMFAML. It shows that OT coupling used for clustering is feasible. Furthermore, our method significantly outperforms WSSC, whose weighting matrix is equivalent to the cost matrix in OT. That again verifies the superiority of OT for clustering over the subspace clustering methods. Compared with the remainder subspace clustering methods, i.e., LRR, LRSC, SSC, and S^4C, the proposed method makes overwhelming effect gaps between them. For UBL and RMMF, with huge mechanism differences from GCOT, they just reach 78% and 85% OA of our proposed method.

On the Indian Pines dataset, there are similar conclusions to the Pavia University dataset, except that EGCSC and NMFAML surpass GCOT for a weak advantage leaving extremely small OA gaps of 0.79 and 1.04 absolute percentage between them, respectively. Overall, our approach achieves comparable performance to the state-of-the-art methods in all the metrics and gains much larger benefits than the other clustering approaches.

For deeper understanding of the clustering results, we analyze the potential reasons behind the improvement derived from GCOT over the other similar methods, in which LRR and LRSC use LRR to form affinity matrices, while SSC, S^4C, WSSC, and EGCSC achieve this by sparse representation. The optimal coupling matrix of GCOT that simultaneously holds the sparsity and low-rank property tends to have more advantage than which contains a single property. The subspace-clustering-based methods (i.e., SSC, S^4C, WSSC, and EGCSC) could cause full zeros of self-representation rows or columns that would hinder clustering, but GCOT does not encounter this problem because of the rows' sum and columns' sum of its coupling amenable to uniform probability density. Moreover, the graph embedding of GCOT using OT probability may better capture the intrinsic geometry of data than that of the EGCSC adopting canonical Euclidean distance. As WSSC characterizes nonlinear data manifolds by the weighting matrix that has similar function to the coupling of GCOT, it does not get rid of the subspace clustering framework, further showing that GCOT is more robust to nonlinear data than the subspace-clustering-based methods.

Although the proposed unsupervised method has achieved remarkable results, even comparable to the supervised manners, it remains trivial because the clustering performance heavily hinges on choosing postprocessing procedures and their corresponding thresholds of the spectral clustering. Besides, the spectral clustering requires inputting entire data at once, which seriously hinders the affinity matrix from fitting large-scale HSIs.

2) Stability of Effect: The clustering performance of the proposed method involves two most critical parameters, i.e., the weight of the entropic regularization ϵ for trade-off sparse transport maps and the number of top k values of each row and column of the OT coupling matrix. As discussed in Section III-B and shown in Fig. 2, the optimal coupling that the clustering performance hinges on gradually diverges with increase in the entropic trade-off ϵ, which is a good property to guide us to find the appropriate parameters. To achieve this, three steps are conducted to search $\epsilon \in (0, +\infty)$. First, we perform a rough search with a step size of 1 until appearing successive poor clustering accuracies or local optimum in a large range, resulting in upper bound 1, 1, and 5 for the Indian Pines, Salinas, and Pavia University datasets, respectively. Second, setting smaller step sizes of 0.01, 0.01, and 0.05 for the three datasets, we find the optimal ϵ and their lower bound of 0.03, 0.03, and 0.5 from 0 to their upper bound. Third, around the optimal values, we search for possibly better parameters with smaller step sizes. The same strategy is executed to find k. According to (8)–(10), when k is set to 0, the graph convolution will degenerate into the self-loop operator and the graph embedding will revert to the original feature that is a benign lower bound for clustering. Theoretically, determining k hinges on the sparsity and intersecting clusters of HSI manifolds, but there are no explicit formulas to calculate it. Hence, combining the rough search with the fine search is a desirable strategy for new datasets.

To investigate the stability of clustering effects, we draw 3-D OA surfaces with respect to parameters ϵ and k for all the datasets. Specifically, we set parameter axis

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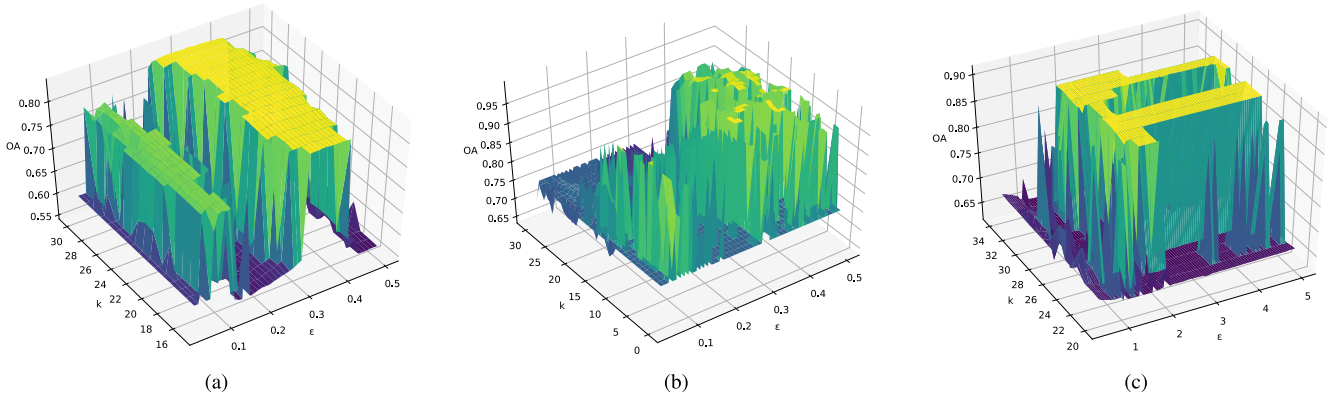


Fig. 4. OA surfaces with respect to ϵ and k to show performance stability. (a) Indian Pines. (b) Salinas. (c) Pavia University.

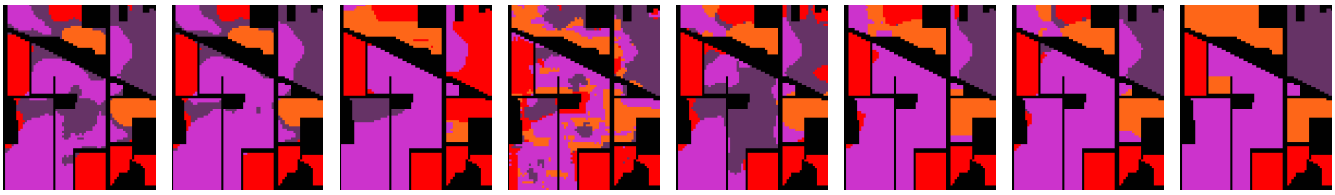


Fig. 5. Resulting segmentation maps of the comparison methods on the Indian Pines dataset. (a) LRR. (b) LRSC. (c) SSC. (d) WSSC. (e) S^4C. (f) EGCSC. (g) GCOT. (h) Ground truth.

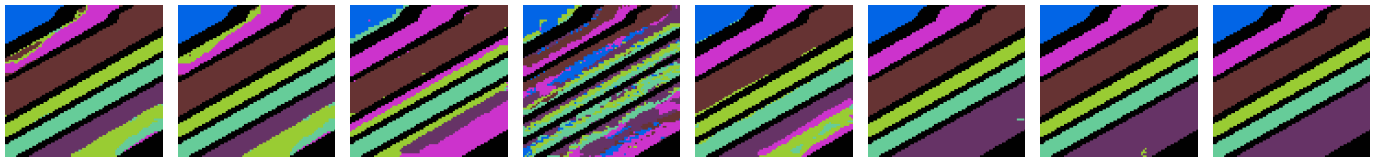


Fig. 6. Resulting segmentation maps of the comparison methods on the Salinas dataset. (a) LRR. (b) LRSC. (c) SSC. (d) WSSC. (e) S^4C. (f) EGCSC. (g) GCOT. (h) Ground truth.

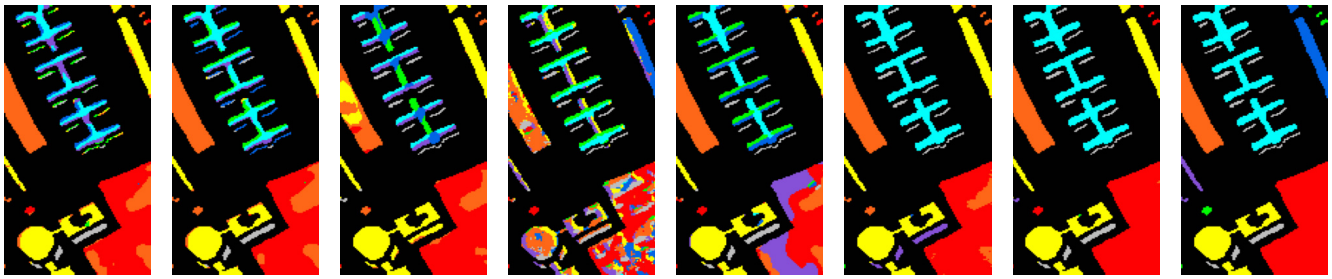
with $\epsilon \in [0.03, 0.5, 0.01]$, $k \in [15, 30, 1]$ for Indian Pines, with $\epsilon \in [0.03, 0.5, 0.01]$, $k \in [1, 30, 1]$ for Salinas, and with $\epsilon \in [0.5, 5, 0.05]$, and $k \in [20, 34, 1]$ for Pavia University, where the triad $[a, b, c]$ denotes a series from a to b in steps of c. Then, the GCOT program traverses the OA values on each grid of parameters to generate the surface of clustering performance. We illustrate the surface in Fig. 4, where the color fades from blue to yellow corresponding to the gradual increase in OA from the minimum to the maximum.

As can be seen from Fig. 4, for the Indian Pines dataset, the best performance appears in a rectangular area in the coordinate system. The parameter ϵ has a great influence on the clustering effects because at any point on the k-axis the cross section of the surface in the ϵ direction forms a zigzag curve. On the contrary, by fixing ϵ and changing k, we find the response values are maintained near a certain value, with a small disturbance. Continuing to analyze the Salinas dataset, we note that the poor performance and better performance are distributed on the top left and right of the surface. The oscillating surface reflects the clustering effects are sensitive to the parameters. Remarkably, the experiment on the Pavia University dataset shows almost the exact opposite conclusion to Indian Pines. That is, the OA values rarely vary with changing ϵ, especially in the highlight areas of better clustering performance on the surface. For the parameter k, the clustering effects, however, are relatively sensitive, resulting in the remarkable scores concentrated around at points 23 and 30.

Based on the observation and analysis, we conclude the two parameters together determine the clustering performance and there are appropriate parameter ranges to ensure the stability of clustering effects. In general speaking, the graph embedding and the entropic regularization of OT contribute positively to learning the coupling matrix for clustering. In addition, we note that the lower bounds of all the effects in Fig. 4 are comparable with the results yielded by most of the methods in Table II.

3) Analysis of Visualization: To visualize the clustering effects and their underlying mechanisms, we plot segmentation maps, transport maps and affinity matrices, and graph embeddings for all the datasets. First, Figs. 5–7 show the segmentation maps by rearranging the clustering labels into map images with the same shapes as the original images.

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(a) (b) (c) (d) (e) (f) (g) (h)

Fig. 7. Resulting segmentation maps of the comparison methods on the Pavia University dataset. (a) LRR. (b) LRSC. (c) SSC. (d) WSSC. (e) S^4C. (f) EGCSC. (g) GCOT. (h) Ground truth.

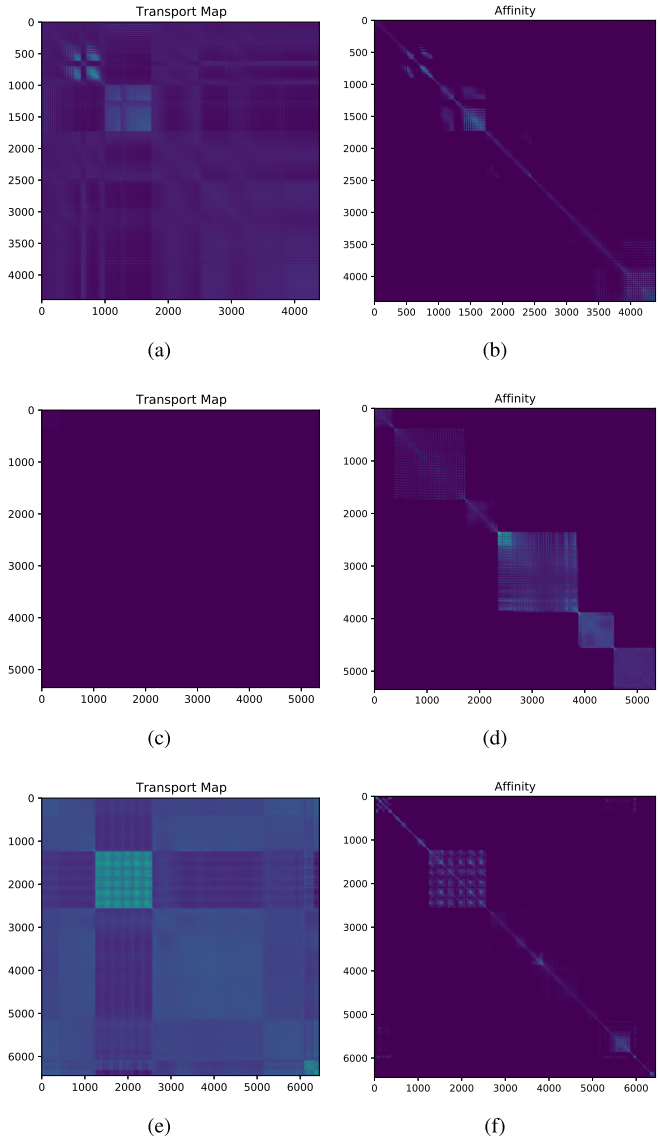


Fig. 8. Illustration of the optimal transport coupling and affinity matrix. (a) and (b) Indian Pines dataset. (c) and (d) Salinas dataset. (e) and (f) Pavia University dataset.

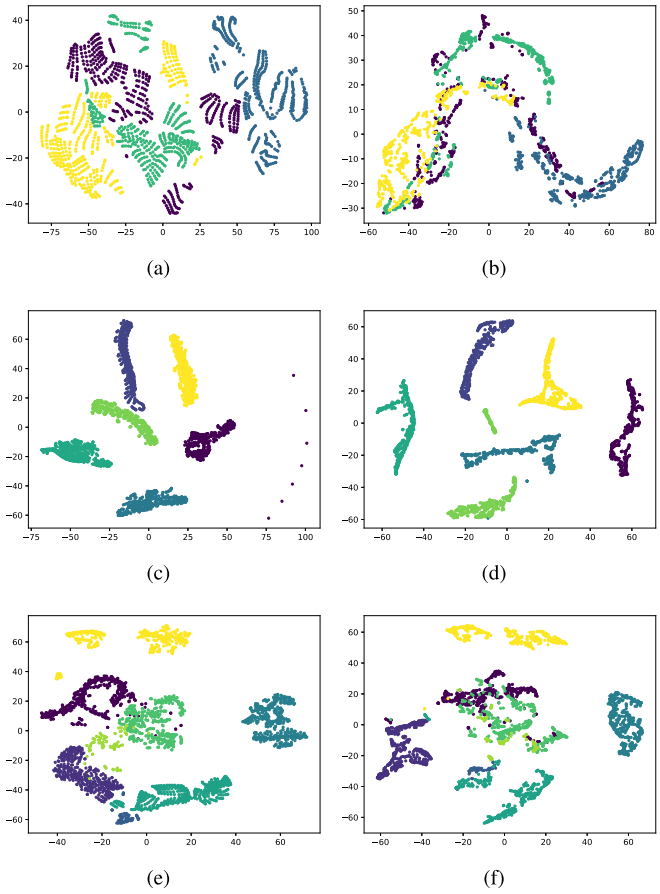


Fig. 9. Illustration of the effects of graph convolution, where (a), (c), and (e) show the original features of the Indian Pines, Salinas, and Pavia University datasets, and (b), (d), and (f) show their graph embedding versions.

Note that because the implementation of UBL, RMMF, and NMFAML is not open source, their segmentation maps are not compared with other methods in the figure. From the experiment on Pavia University of Fig. 7, GCOT achieves the best map that contains very few stray colors and is closest to real segmentation. Similar segmentation maps are obtained by GCOT and EGCSC on Indian Pines in Fig. 5 and on Salinas in Fig. 6, having clearer boundaries than other methods. These conclusions are consistent with the objective evaluation.

Then, Fig. 8 shows the transport maps, i.e., in the left figures, and their corresponding affinity matrices, i.e., in the right figures of (a), (c), and (e). The clearer the diagonal block structure of the affinity matrix, the better the clustering results are obtained. It is exactly reflected in the figures of (b), (d), and

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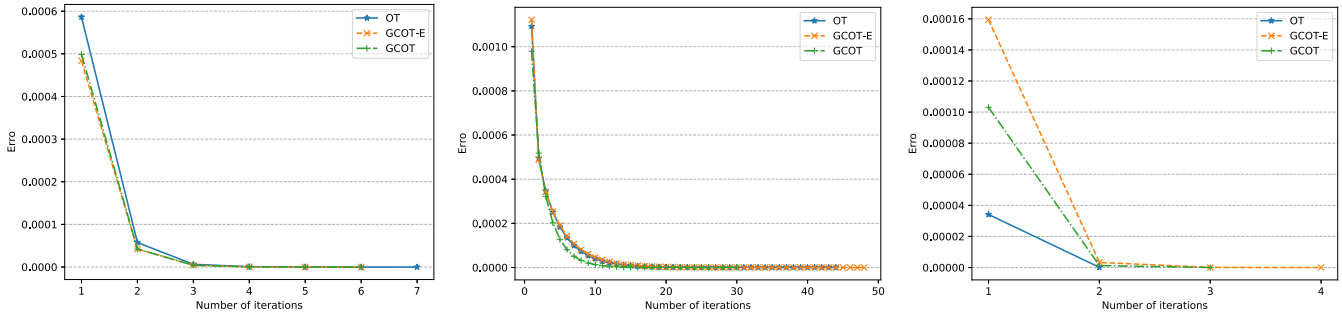


Fig. 10. Iterations of different OT problems to convergence on (a) Indian Pines, (b) Salinas, and (c) Pavia University datasets.

TABLE III
COMPARISON OF TIME CONSUMPTION (SEC) OF DIFFERENT CLUSTERING ALGORITHMS ON THE INDIAN PINES, SALINAS, AND PAVIA UNIVERSITY DATASETS

Data sets	LRR [12]	LRSC [13]	SSC [10]	WSSC [65]	S^4C [15]	EGCSC [27]	GCOT
Indian Pines	32.889	12.057	138.093	450.532	500.895	39.016	28.475 (0.871 + 1.165 + 0.909 + 25.53)
Salinas	56.238	23.053	202.741	764.350	850.482	73.559	56.462 (4.599 + 2.179 + 3.7 + 45.984)
Pavia University	80.949	40.035	311.913	1199.064	1330.285	74.789	69.058 (1.864 + 2.774 + 1.932 + 62.488)

(f). In Fig. 8(c), due to the divergent transmission probabilities, it is difficult to see their transmission path visually, but the desired sharp blocks, nevertheless, occur in the affinity matrix after applying normalization.

Finally, the impacts of graph convolution for data feature are shown in Fig. 9. Concretely, for each dataset, 400 points are sampled from the original features and graph embeddings in each cluster and reduced into the 2-D plane by t-distribution stochastic neighbor embedding (t-SNE). The clusters are distinguished in different colors. As we have seen, inside the clusters, the graph embeddings [the right figures of (b), (d), and (f)] are more compact than the original features [the left figures of (a), (c), and (e)]. Through observing Fig. 9(c) and (d), the graph convolution shows the ability that smooths discrete points in a cluster to enforce them closer together, which is beneficial to clustering. Especially, there are 7 distributed outlying entries on the right side of (c) which have been concentrated in an expected cluster in Fig. (d). In contrast, it is worth noting that there are more or less overlaps between different clusters in Fig. (b) and (f) compared with their left figures. Despite all this, the intradistances between the embeddings of the same clusters are holistically smaller than the interdistances between that of the different clusters in the right cases, which exactly meets the applicability condition of our model as shown in Theorem 1. Because the points of each cluster in Fig. (a) are more dispersed than the counterpart in Fig. (b) and the distances between points in different clusters are relatively average, the latter ratios of the interdistances and intradistances are larger than the former ones.

4) Comparison of Time Consumption: For evaluating the efficiency of the proposed method, we count the time consumption for several algorithms setting the maximum iterations to 200. The comparison results are summarized in Table III. Since having a closed-form solution, LRR gives the best efficiency performance but poor clustering effects. Among all the state-of-the-art methods, the efficiency of GCOT outperforms EGCSC and other algorithms, owing to the highly efficient optimization of the entropically regularized OT. Its total time consumption consists of four parts, including solving optimal coupling of OT on original feature, constructing adjacent graph matrix, solving optimal coupling of GCOT on graph embedding, and spectral clustering. The spectral clustering consumes about 90% of the total time, while GCOT takes only a few seconds from the first three steps to generating the affinity matrix.

C. Ablation Study

1) Effect of Different Graphs for GCOT: To verify the improvement brought by the graph embedding, we conduct comparison experiments on entropically regularized OT, GCOT-E, and GCOT. Note that the basic OT directly learns the optimal transport coupling on the original feature, while GCOT-E replaces the OT probability of GCOT with the Euclidean distance to yield the adjacent matrix of the graph. From the clustering results shown in Table IV, we arrive at the following tendencies. First, the basic OT produces remarkable results better than most of the other clustering methods in Table II and is comparable to the frontier methods. Second, the performance of GCOT-E and GCOT indicates that graph embedding is helpful to improve the clustering accuracy. Third, the best scores achieved by GCOT show that the transport probability between data features can capture more local geometric structure information than the sought after Euclidean distance used in k-nearest neighbor (k-NN) for building graphs.

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TABLE IV
ABLATION STUDY FOR GCOT WITH DIFFERENT GRAPHS

Methods	Indian Pines	Salinas	Pavia University
OT	0.8285	0.9974	0.8876
GCOT-E	0.8363	0.9974	0.8923
GCOT	**0.8404**	**0.9979**	**0.9102**

2) Convergence Comparison Between Different OT: As seen in Fig. 10, all the OT variants tend to converge within ten iterations on the Indian Pines and Pavia University datasets. GCOT converges at steps 2 and 30 on the Indian Pines and Salinas datasets and achieves the best efficiency performance against other methods. On the Pavia University dataset, GCOT just converges in three iterations. Compared with OT and GCOT-E, the proposed method maintains a good balance between their efficiency.

V. CONCLUSION

We propose a novel OT defined on graph convolution (i.e., GCOT) for HSI spectral clustering. For forming the affinity matrix of spectral clustering by OT coupling, we provide theoretical proofs that explain the coupling matrix has sparsity and low-rank property. Moreover, we further analyze the general form of GCOT and give the relationship with subspace clustering. The experimental results indicate the superiority of our method on three datasets in terms of three objective metrics. The visualization of the intermediate results of clustering shows consistent conclusions with the objective evaluation. Especially, among all the frontier methods, our method achieves the best efficient performance.

Although the proposed OT method applied on the graph embedding can capture the intrinsic geometric structure of data, these two steps are relatively independent without consistency. In the future work, we consider integrating them with spectral-spatial features of data into a coclustering framework without feature preprocessing procedures.

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Shujun Liu (Graduate Student Member, IEEE) received the M.S. degree in computer software and theory from the Chengdu University of Technology, Chengdu, China, in 2017, where he is currently pursuing the Ph.D. degree in Earth exploration and information technology.

His research interests lie in machine learning theory and hyperspectral image processing.



Huajun Wang received the B.S. degree from Beijing Normal University, Beijing, China, in 1986, and the M.S. and Ph.D. degrees in electronic science and technology from the University of Electronic Science and Technology of China, Chengdu, China, in 1997 and 2001, respectively.

He was a Post-Doctoral Researcher with the University of Electronic Science and Technology of China, from 2000 to 2004, and also with Lomonosov Moscow State University, Moscow, Russia. Since 2005, he has been a full Professor with the Chengdu University of Technology, Chengdu. His research interests include machine learning, remote sensing, and space physics.

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